

Federated Diabetic Retinopathy Grading Across Rural Screening Centres Without Sharing Patient Images

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TRACK T3 · ARTIFICIAL INTELLIGENCE & APPLIED ML

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1 PROBLEM STATEMENT

Rural screening camps capture retinal images but have no ophthalmologist, so grading waits weeks. A shared model would close that gap, but hospital policy forbids patient images leaving the centre that captured them, and no single centre holds enough images to train alone. Camera models differ between centres, so a model trained at one performs poorly at another.

2 OBJECTIVES

- Train a five-class grading model across six centres without moving a single image.
- Quantify the accuracy gap against centralised training on the pooled dataset.
- Handle non-IID data — differing camera models and grade distributions per centre.
- Keep communication under 20 MB per round so a 4G link at a camp suffices.

3 METHODOLOGY

- Data partition — 11,400 fundus images, six centres, non-IID by camera
- Local training — EfficientNet-B0, three local epochs per round
- Aggregation — FedAvg with server-side momentum, 40 rounds
- Drift correction — batch-norm statistics kept local, never averaged
- Evaluation — held-out centre, quadratic weighted kappa

4 EXPERIMENTAL SETUP

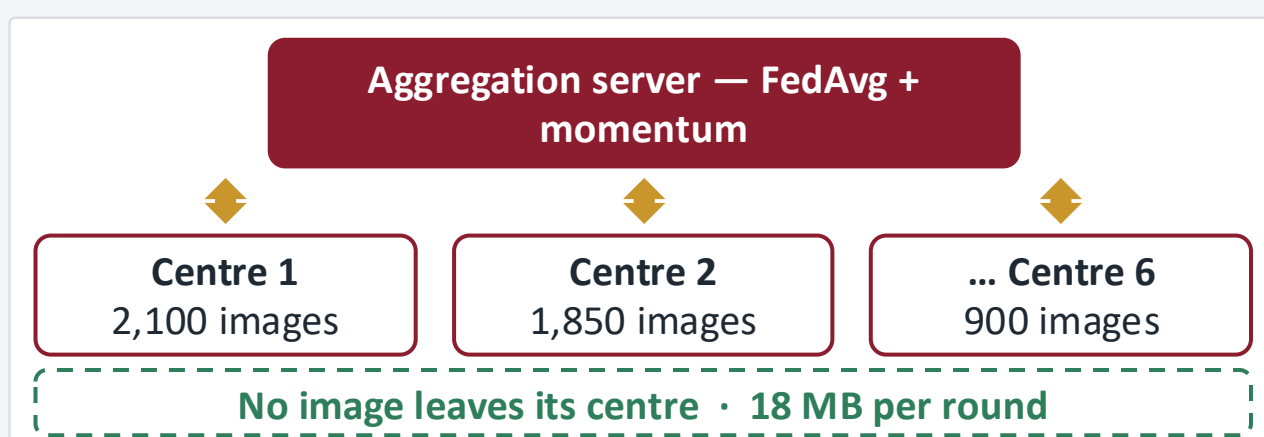


Fig. 1 — Federated setup. Each centre trains locally and returns only weight updates; the batch-norm statistics stay behind, which is what lets the aggregated model transfer across cameras.

5 RESULTS & DISCUSSION

0.847

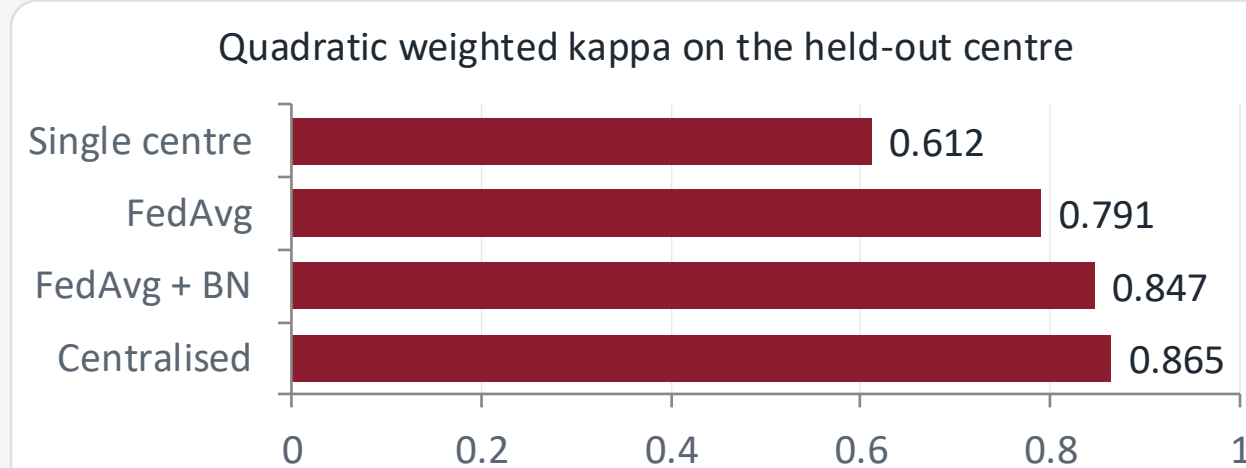
QUADRATIC
WEIGHTED KAPPA

1.8 pts

GAP TO CENTRALISED
TRAINING

18 MB

PER
COMMUNICATION
ROUND



Keeping batch-norm statistics local is worth 5.6 points of kappa — it is the camera-to-camera colour difference, not the grade imbalance, that breaks plain FedAvg. The residual 1.8-point gap to centralised training sits almost entirely in grade 2, the class with fewest examples at every centre.

6 NOVELTY & OUR OWN CONTRIBUTION

- Non-IID partition built from real camera metadata across six centres, not a synthetic Dirichlet split.
- Keeping batch-norm statistics local recovers 5.6 kappa points; the cause is camera colour, not label skew.
- The 18 MB per-round budget achieved by gradient sparsification tuned by me over seven sparsity levels.
- Reused and cited: the FedAvg algorithm [1], EfficientNet-B0 [3] and the weighted kappa protocol [4].

7 CONCLUSION & IMPACT

Federated training reaches 0.847 kappa on a centre it has never seen, within 1.8 points of pooling every image centrally, while no patient image leaves the centre that captured it. Six camps could share one grading model over a 4G link. Next: prospective validation at two live camps and a grade-2 oversampling study.

SDG 3

SDG 10

Good Health & Well-being · Reduced Inequalities

8 REFERENCES (IEEE style, max 6)

- [1] B. McMahan et al., "Communication-efficient learning of deep networks from decentralized data," Proc. AISTATS, pp. 1273–1282, 2017.
- [2] P. Kairouz et al., "Advances and open problems in federated learning," Found. Trends Mach. Learn., vol. 14, 2021.
- [3] M. Tan and Q. Le, "EfficientNet: rethinking model scaling for CNNs," Proc. ICML, pp. 6105–6114, 2019.
- [4] J. Cohen, "Weighted kappa," Psychological Bulletin, vol. 70, no. 4, pp. 213–220, 1968.